Wentao Zhang
Research Assistant, Power Electronics and Machines Centre
Faculty of Engineering, University of Nottingham
wentao.zhang@nottingham.ac.uk · (+44) 7402 114 543

12 May 2026

Professor Markus Barth
School of Electrical Engineering and Computer Science
Faculty of Engineering, Architecture and Information Technology
The University of Queensland
St Lucia QLD 4072, Australia

Dear Professor Barth,

Application for Lecturer in Electrical Engineering / Biomedical Engineering — R-63679

I am writing to apply for the Lecturer position in Electrical Engineering / Biomedical Engineering (R-63679) advertised by the School of Electrical Engineering and Computer Science. I am currently a Research Assistant at the Power Electronics and Machines Centre (PEMC), University of Nottingham, having completed my PhD in Electrical Engineering at Southeast University, Nanjing, in September 2025. The balance the role offers between teaching across the electrical-engineering curriculum and a research home in a School with strong Imaging, Sensing and Biomedical Engineering activity is an excellent fit for the methodology-led programme I want to build at UQ.

My research develops multiphysics modelling and AI-driven surrogate techniques for the analysis, design and optimisation of magnetic and electromagnetic devices. The work has three methodological strands: (i) strongly coupled electromagnetic–thermal–fluid–mechanical finite-element analysis; (ii) high-resolution lumped-parameter thermal networks combined with oil-cooling fluid–solid solvers; and (iii) operator-learning neural surrogates for the magnetic field that achieve design-loop accuracy at orders-of-magnitude lower cost than conventional numerical analysis. To date I have authored 22 journal papers (seventeen published IEEE Transactions, one in early access), eighteen peer-reviewed conference papers, and hold nine patents. The work has been recognised by the 2025 1st Prize in the IEEE TEC Electric Vehicle Design Competition, the 2025 2nd Prize Paper Award of *IEEE Transactions on Transportation Electrification*, the 1st Prize in the IEEE IAS CMD Thesis Contest, and three best-paper awards at IEEE-affiliated conferences. My principal application domain has been high-power-density electric drive systems for electrified transport; the underlying machinery — magnetic-field forward modelling, multiphysics coupling and learned surrogates — is application-agnostic and well placed to support cross-disciplinary collaboration within the School.

I am keen and prepared to teach into several of the curriculum areas listed in the position description, particularly **introductory programming for engineering students**, **embedded and digital systems** (introductory through advanced undergraduate), and **electrical-engineering team and individual capstone projects**. My day-to-day toolchain — Python, MATLAB/Simulink, ANSYS, and embedded firmware on TI C2000 and STM32 — supports both contemporary curriculum design and the project-based components students value most. During my PhD I served as a teaching assistant and laboratory demonstrator at Southeast University and informally co-mentored junior PhD and MSc students in Prof. Wei Hua's group, evidenced in the co-authorship pattern of my publication record. I would welcome UQ's standard induction support for new Level B Lecturers as I take on full course-coordination responsibility.

Beyond research and teaching, I am Co-Organiser and Co-Chair of a Special Session at the 2026 International Conference on Electrical Machines (ICEM 2026, Covilhã, Portugal) on *Next-Generation Electrified Transport: Advances in Electric Drive Systems*, jointly with Dr Lingyun Shao (NUAA) and collaborators across China, the UK and the EU. I would welcome the opportunity to develop similar collaborative initiatives with the School and across the wider Faculty.

I have attached my CV and detailed responses to the “About You” selection criteria. Thank you for your consideration; I would be very pleased to discuss how I could contribute to the School’s teaching programme and research strategy.

Yours sincerely,

Wentao Zhang